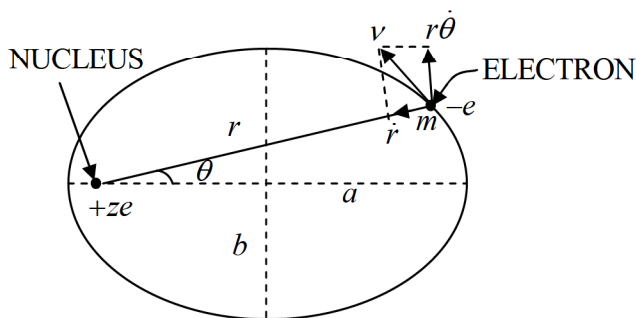


Chapter Three

Sommerfeld extension of Bohr model

1.4 Sommerfeld extension of Bohr model

In Bohr model of hydrogen atom, the electron could move only in circular orbits. But Sommerfeld produce a advance model in which the electron revolved about proton in elliptical orbit where proton is at one of foci of ellipse. The model is exactly analogical to Kepler's model of sun and earth system.



From Sommerfeld quantization rule for radial direction $\oint p_r dr = n_r h$

From Sommerfeld quantization rule for azimuthal direction $\oint p_\theta d\theta = n_\theta h$

So $p_\theta = \frac{n_\theta h}{2\pi}$

We know that radial momentum $p_r = m\dot{r}$ and angular momentum $p_\theta = mr^2\dot{\theta}$

So polar equation of ellipse is given by $\frac{l}{r} = 1 + \varepsilon \cos \theta$ where l is latus rectum and ε is eccentricity.

Differentiating both sides, we will get

$$-\frac{l}{r^2} \dot{r} = -\varepsilon \sin \theta \dot{\theta} \Rightarrow \frac{l}{r^2} \dot{r} = \varepsilon \sin \theta \frac{p_\theta}{mr^2} \Rightarrow \dot{r} = \varepsilon \sin \theta \frac{p_\theta}{m} \Rightarrow p_r = m\dot{r} = p_\theta \varepsilon \sin \theta$$

$$-\frac{l}{r^2} \dot{r} = -\varepsilon \sin \theta \dot{\theta} \Rightarrow \frac{l}{r^2} \frac{dr}{dt} = \varepsilon \sin \theta \frac{d\theta}{dt} \Rightarrow dr = \frac{\varepsilon \sin \theta d\theta}{(1 + \varepsilon \cos \theta)^2}$$

$$\oint p_r dr = p_\theta \int_0^{2\pi} \varepsilon^2 \frac{\sin^2 \theta d\theta}{(1 + \varepsilon \cos \theta)^2} = n_r h$$

$$2\pi p_\theta \left(\frac{1}{\sqrt{1 - \varepsilon^2}} - 1 \right) = n_r h \Rightarrow 2\pi \frac{n_\theta h}{2\pi} \left(\frac{1}{\sqrt{1 - \varepsilon^2}} - 1 \right) = n_r h$$

$$\sqrt{1 - \varepsilon^2} = \frac{b}{a} = \frac{n_\theta}{n_r + n_\theta} \Rightarrow \frac{n_\theta}{n} = \frac{b}{a} \text{ where } n_\theta = 1, 2, 3 \dots, n_r = 0, 1, 2, 3 \dots \text{ and } n = 1, 2, 3 \dots$$

(If $n_\theta = 0$ then $b = 0$ which is not support the basic property of ellipse so $n_\theta \neq 0$)

Sommerfeld also calculated the other parameter of Hydrogen atom

Energy

$$F = \frac{mz^2 e^4 \left(\frac{b}{a} \right)^2}{32\pi^2 \epsilon_0^2 p_\theta^2} \quad \frac{b}{a} = \frac{n_\theta}{n} \text{ and } p_\theta = \frac{kh}{2\pi} \text{ so } F = -\frac{mz^2 e^4}{8\epsilon_0^2 h^2} \left(\frac{1}{n^2} \right) \Rightarrow -\frac{mz^2}{2h^2} \left(\frac{e^2}{4\pi \epsilon_0} \right)^2$$

Size and shape of Sommerfeld's Elliptical orbits

$$\text{Semimajor Axis } a = \frac{n^2 h^2 \epsilon_0}{\pi m z e^2} = \frac{4\pi \epsilon_0}{m z e^2} n^2 \Rightarrow \frac{n^2}{z} a_0$$

$$\text{Semi minor axis } b = \frac{nn_\theta}{z} a_0$$

Possible shape of orbit

$n = 1, n_\theta = 1, a = a_0, b = a_0$ which is circular in nature with radius a_0

$n = 2, n_\theta = 2, a = 4a_0, b = 4a_0$ which is circular in nature with radius $4a_0$

$n = 2, n_\theta = 1, a = 4a_0, b = 2a_0$ which is elliptical in nature.

1.5 Summary of solution Schrödinger wave equation for hydrogen atom with (this topic is detail discuss in quantum mechanics course)

The hydrogen atom consist of heavy essentially motionless proton (which can be put it at origin)of charge e together with a much lighter electron (charge $-e$) that orbits around

it ,bound by mutual attraction from coulomb potential as $V(r) = \frac{-e^2}{4\pi \epsilon_0 r}$.the coulomb

potential is central potential so angular momentum of electron is conserve .so angular momentum operator $\hat{L}^2$ will commute with Hamiltonian . $[H, \hat{L}^2] = 0$ If $\psi(r, \theta, \phi)$ is solution of Schrodinger wave equation then $\psi(r, \theta, \phi) = R(r) Y_l^m(\theta, \phi)$

Where $R(r)$ is radial wave function and $Y_l^m(\theta, \phi)$ is spherical harmonics.

The Schrodinger equation reduce to

$$\frac{-\hbar^2}{2m} \left[\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} \right] + \left[\frac{l(l+1)\hbar^2}{2mr^2} + V(r) \right] R(r) = ER(r) \text{ where } V(r) = \frac{-e^2}{4\pi \epsilon_0 r}$$

Putting $R(r) = \frac{u(r)}{r}$ the above equation reduces

$$\frac{-\hbar^2}{2m} \frac{d^2 u}{dr^2} + \left[\frac{l(l+1)\hbar^2}{2mr^2} - \frac{e^2}{4\pi \epsilon_0 r} \right] u(r) = Eu(r) \Rightarrow \frac{-\hbar^2}{2m} \frac{d^2 u}{dr^2} = \left[E - \frac{l(l+1)\hbar^2}{2mr^2} + \frac{e^2}{4\pi \epsilon_0 r} \right] u(r)$$

The general solution is given

$$\psi(r, \theta, \phi) = L_n^l(r) Y_l^m(\theta, \phi) \exp\left(-\frac{r}{na}\right) \text{ Where } L_n(r) \text{ is Laguerre Polynomial}$$

Where $n = 1, 2, 3, \dots$ and $l = 0, 1, 2, \dots, n-1$ can be found from $j_{\max} + l + 1 = n$ if $j_{\max} = 1$

$m = -l, l+1, -l+2, \dots, 0, 1, 2, \dots, l-1, l$ then $l = n-2$

Where $L_n(r)$ is Laguerre Polynomial

$$L_n(r) = \exp(r) \left(\frac{d}{dx} \right)^n (\exp(-r) r^n) \quad L_{n-l}^l(r) = (-1)^l \left(\frac{d}{dr} \right)^l L_n(r)$$

$$\text{The general solution is given by } \psi_{n,l,m} = \sqrt{\left(\frac{2}{na} \right)^3 \frac{(n-l-1)!}{2n(n+1)!}} \exp\left(-\frac{r}{na}\right) \left[L_{n-l-1}^{2l+1} \left(\frac{2r}{na} \right) \right] Y_l^m(\theta, \phi)$$

The first few radial wave functions of hydrogen atom

$$R_{10} = 2a^{-3/2} \exp\left(-\frac{r}{a}\right) \text{ for } n=1 \text{ and } l=0$$

$$R_{20} = \frac{1}{\sqrt{2}} a^{-3/2} \left(1 - \frac{1}{2} \frac{r}{a}\right) \exp\left(-\frac{r}{2a}\right)$$

$$R_{21} = \frac{1}{\sqrt{24}} a^{-3/2} \frac{r}{a} \exp\left(-\frac{r}{2a}\right)$$

Ortho normal relation is given by $\int_0^{\infty} R_n^* R_n r^2 dr = \delta_{n'n}$

Some important spherical harmonics

$$|0,0\rangle = Y_0^0 = \frac{1}{\sqrt{4\pi}}, |1,0\rangle = Y_1^0 = \sqrt{\frac{1}{8\pi}} \sin \theta e^{i\phi}, |1,-1\rangle = Y_1^{-1} = \sqrt{\frac{3}{8}} \sin \theta e^{-i\phi}$$

Orthonormal relation of $Y_l^m(\theta, \phi)$ $\int_0^{2\pi} \int_0^{\pi} Y_l^{m'}(\theta, \phi) Y_l^m(\theta, \phi) \sin \theta d\theta d\phi = \delta_{l'l} \delta_{m'm}$

which can be represented as $\langle l', m' | l, m \rangle = \delta_{l'l} \delta_{m'm}$

Orthonormal relation for total wave function

$\psi_{n,l,m}(r, \theta, \phi)$ where n is principle quantum number, l is orbital quantum number and m is magnetic quantum number.

$$\int_0^{\infty} \int_0^{\pi} \int_0^{2\pi} \psi_{n',l',m'}^* \psi_{n,l,m} r^2 \sin \theta dr d\theta d\phi = \delta_{n',n} \delta_{l',l} \delta_{m',m}$$

Effective potential, ground state energy and ionization energy

Effective potential is given by $V_{\text{effective}} = \frac{\hat{L}^2}{2mr^2} - \frac{e^2}{4\pi\epsilon_0 r}$ is shown in fig.

The bound state energy is negative which is given by $E_n = -\frac{m}{2\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0} \right) \frac{1}{n^2}$

Ground state energy is minimum energy of electron in bound state which is given by

$$E_1 = -\frac{m}{2\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0} \right) = -13.6 \text{ eV}.$$

Ionization energy is the minimum energy that an electron in a hydrogen atom or has to absorb to come out of the influence of the electrostatic potential. If E_s is ionization energy is

$E_s = E_f - E_i$. Where E_f is minimum final energy when electron will unbounded i.e. $E_f = 0$. E_i is minimum initial energy when it is bounded i.e. $E_i = -13.6 \text{ eV}$ $E_s = 0 - (-13.6 \text{ eV}) = 13.6 \text{ eV}$.